

Typhoidal Tularemia: A Comprehensive Disease Characterization Report

Disease: Typhoidal Tularemia **MONDO ID:** MONDO:0000321 **Category:** Infectious Disease
Causative agent: *Francisella tularensis* (NCBI:txid263) **Report date:** 2026-09-25 | **Iterations completed:** 5 | **Confirmed findings:** 14 | **Papers reviewed:** 67 **Report type:** Literature-based synthesis (no primary dataset). Evidence drawn from human clinical case reports/series, epidemiologic surveillance, in vitro cell biology, and animal (mouse/rat/*Drosophila*) models, as indicated per claim.

Summary

Typhoidal tularemia is the severe, **systemic (septicemic) clinical form** of tularemia, a zoonosis caused by the Gram-negative, facultative intracellular coccobacillus *Francisella tularensis*. It is one of six recognized clinical forms (ulceroglandular, glandular, oculoglandular, oropharyngeal, typhoidal, and pneumonic) and is defined by an acute typhoid-like febrile illness — high fever, chills, malaise, anorexia, prostration — that occurs **without a prominent inoculation ulcer or regional lymphadenopathy**. Because it lacks these localizing signs, it is frequently mistaken for typhoid/enteric fever, sepsis of unknown origin, or hematologic malignancy, causing diagnostic delay. Together with the pneumonic form, typhoidal tularemia carries the **highest case-fatality of all tularemia presentations — up to ~60% if untreated** (PMID: 40107886).

Mechanistically, the disease is driven entirely by the pathogen and host innate immunity — there is **no human genetic cause**. Low-dose exposure (inhalation of ≤ 10 organisms can cause lethal disease) leads to macrophage uptake, where the *Francisella* Pathogenicity Island (FPI) proteins IgIC/IgID and the master regulator MglA/SspA mediate escape from the phagosome into the cytosol, followed by rapid intracellular replication and dissemination through the reticuloendothelial system (spleen, liver, lymph nodes, bone marrow, lung). The host counters with the AIM2 inflammasome (sensing cytosolic bacterial DNA $\rightarrow$ caspase-1 $\rightarrow$ IL-1 β /IL-18 $\rightarrow$ pyroptosis) and IFN- γ -mediated macrophage activation. An upstream virulence trait — an unusually under-acylated, hypo-phosphorylated lipid A — allows *F. tularensis* to evade early TLR4/MD-2 recognition, delaying protective inflammation.

Clinically, the disease is diagnosed primarily by **serology** (microagglutination/ELISA), supplemented by blood culture (more often positive in systemic disease), PCR, and increasingly cell-free DNA/metagenomic sequencing. First-line treatment for severe/typhoidal disease is the aminoglycoside **gentamicin**, with fluoroquinolones (ciprofloxacin, levofloxacin) and tetracyclines (doxycycline) as alternatives; bacteriostatic tetracyclines carry higher relapse risk. **No licensed vaccine** exists, so prevention rests on exposure avoidance, vector control, and post-exposure antibiotic prophylaxis. Prognosis is excellent with prompt appropriate therapy but poor when treatment is delayed. This report characterizes the disease across all 15 template sections, mapping findings to ontology terms and anchoring every claim in primary literature.

Key Findings

F001 — Typhoidal tularemia is a severe systemic form of tularemia caused by *Francisella tularensis*

Typhoidal tularemia is one of six typical clinical pictures produced by *F. tularensis* depending on the route of infection. As the multiple routes of infection "result in six typical clinical pictures (ulceroglandular, glandular, oculoglandular, oropharyngeal, typhoidal, and pneumonic)" (PMID: 40107886), the typhoidal form is distinguished by systemic febrile illness resembling typhoid fever, **without** a prominent inoculation ulcer or regional lymphadenopathy. A documented case illustrates the systemic, hematologic character: a patient "presented with a two-week history of high-grade fever, severe malaise, anorexia, and laboratory evidence of pancytopenia with hypoglycemia," and the "clinical course was complicated by pericarditis" (PMID: 41458397). This confirms the constitutional and reticuloendothelial character of the disease and its capacity for organ complications.

F002 — Typhoidal and pneumonic forms carry the highest fatality (up to 60% untreated)

Case-fatality is dramatically stratified by clinical form and pathogen subspecies. "If not promptly diagnosed and treated, the fatality rate can be as high as 60%, with the poorest outcomes reported in the pneumonic and typhoidal forms" (PMID: 40107886). Severity is

subspecies-dependent: *F. tularensis* subsp. *tularensis* (type A, North America) is more virulent than subsp. *holarctica* (type B, Europe/Asia). Within type A, geographic clades differ — "type A-west infections are less severe than either type B or type A-east infections" (PMID: 16836829).

F003 — Pathogenesis: FPI-encoded IgIC and regulator MglA drive phagosomal escape and cytosolic replication

After macrophage engulfment, the *Francisella*-containing phagosome matures to a LAMP-1/LAMP-2⁺ late-endosomal stage but avoids lysosomal fusion, then is disrupted, releasing bacteria into the cytosol where they replicate. The FPI protein IgIC and its regulator are central: studies identify "the *Francisella* pathogenicity island (FPI) protein IgIC and its regulator MglA in the intracellular fate" (PMID: 15953029). Functional work shows "the IgIC, IgID, and MglA proteins each directly or indirectly critically contribute to the virulence of *F. tularensis* LVS, including its intracellular replication, cytoplasmic escape, and inhibition of acidification of the phagosomes" (PMID: 18474647). Host activation reverses this: nitric oxide donors inhibit *mglA* — "Addition of SNAP led to significantly increased colocalization between LAMP-1 and bacteria, indicating containment of *F. tularensis* in the phagosome within 2 h" (PMID: 21700740).

F004 — Treatment: aminoglycosides (gentamicin) first-line for severe/typhoidal disease

Antibiotic choice is stratified by severity: "Gentamicin is the first-line treatment for severe tularemia, while fluoroquinolones and tetracyclines are commonly the drugs of choice in less severe forms" (PMID: 40107886). Fluoroquinolone regimens perform well in severe respiratory type B disease: among 67 case-patients (median age 66, 81% male), "30-day mortality was 1.5% (1 of 67)," and "one disease relapse occurred with doxycycline treatment" (PMID: 38294118). Bacteriostatic tetracyclines carry higher relapse rates than bactericidal aminoglycosides/fluoroquinolones.

F005 — Epidemiology: Northern Hemisphere zoonosis with type A/B geographic split

Tularemia occurs across North America, Europe, and northern Asia (not the Southern Hemisphere). "Tularaemia has been reported in more than 250 animal species including man" (PMID: 1305858). Two subspecies partition ecologically: type A cycles through cottontail rabbits and ticks (terrestrial), type B through aquatic rodents (muskrats, beaver, voles; water-borne). Large outbreaks occur — in central Sweden in 2019, "a total of 979 cases (734 laboratory-

confirmed) have been reported, mainly from counties in central Sweden" over ~10 weeks (PMID: 31640844). The typhoidal form constitutes roughly 10–14% of cases in temperate series; a Norwegian series reported "glandular (14.4%), typhoidal (14.4%), respiratory (13.3%) and ulceroglandular (12.8%) tularemia" (PMID: 24874046).

F006 — Diagnosis relies on serology; blood culture more often positive in typhoidal disease

Serology is the cornerstone: "Serology is still considered to be a cornerstone in tularemia diagnosis due to the low sensitivity of bacterial culture and the lack of standardization in PCR methodology" (PMID: 20220165). A rapid immunochromatographic test achieved excellent performance — "the ICT had a sensitivity of 98.3% ... and a specificity of 96.5%" (PMID: 20220165). Antibodies typically appear ~2 weeks after onset, limiting early diagnosis. Bacteremia is detectable in systemic disease: "An unexpectedly high number (3.9%) of the patients had positive blood culture with *Francisella tularensis*" (PMID: 24874046). Culture is hazardous (BSL-3), and cell-free DNA/metagenomic sequencing has diagnosed occult typhoidal cases (PMID: 41160772).

F007 — Prevention: no licensed vaccine; exposure avoidance and post-exposure prophylaxis

"No licensed vaccine is available in the prophylaxis of tularemia and this is need of the time and high-priority research area" (PMID: 32989563). The live vaccine strain (LVS) provides partial, route-limited protection but is not routinely licensed. Prevention rests on environmental/animal control, arthropod-bite and contaminated-water/food avoidance, and post-exposure prophylaxis (doxycycline or ciprofloxacin) after high-risk exposure. Bioterrorism preparedness is relevant because "a weapon using airborne tularemia would likely result 3 to 5 days later in an outbreak of acute, undifferentiated febrile illness with incipient pneumonia" (PMID: 11386933).

F008 — Animal models and natural disease

Murine models are the mainstay: "Inhalation of 10 or fewer organisms results in an acute and potentially lethal disease called pneumonic tularemia" (PMID: 28372827). Systemic dissemination and cytokine responses are reproducible — treatment reduced "bacterial burden in the spleen and liver, which correlated with a significant reduction in the pro-inflammatory cytokines IFN- γ , MCP-1, IL-6, and TNF- α " (PMID: 27714591). Naturally, "Type A is

reported to have a terrestrial cycle with the main reservoirs being cottontail rabbits (*Sylvilagus* spp.) and ticks" (PMID: 1305858). Fischer 344 rats and *Drosophila melanogaster* serve as additional models.

F009 — Clinical phenotype: systemic febrile illness with reticuloendothelial signs and cytopenias

Typhoidal tularemia presents acutely (incubation ~3–5 days, range 1–14) with high fever, chills, malaise, anorexia, weight loss, headache, myalgia, and prostration — characteristically without a skin ulcer or regional lymphadenopathy. Systemic dissemination produces hepatosplenomegaly and laboratory abnormalities including pancytopenia, hypoglycemia, and elevated transaminases; complications include pneumonia, pleural effusion, pericarditis/myocarditis, and sepsis (PMID: 41458397). Enteric exposure can add abdominal pain, nausea, vomiting, and diarrhea. Granulomatous (necrotizing) inflammation is typical and can mimic malignancy: CT-guided biopsies "revealed a non-specific necrotizing granulomatous inflammation" in lesions "highly suggestive of malignancy" (PMID: 41482246).

Suggested HPO terms: Fever (HP:0001945), Chills (HP:0025143), Weight loss (HP:0001824), Myalgia (HP:0003326), Headache (HP:0002315), Hepatosplenomegaly (HP:0001433), Splenomegaly (HP:0001744), Pancytopenia (HP:0001876), Hypoglycemia (HP:0001943), Elevated circulating hepatic transaminase (HP:0002910), Pericarditis (HP:0001701), Pneumonia (HP:0002090), Sepsis (HP:0100806), Abdominal pain (HP:0002027), Diarrhea (HP:0002014).

F010 — Cytosolic escape triggers the AIM2 inflammasome as a key innate defense

Once in the cytosol, bacteriolysis releases dsDNA sensed by AIM2. "AIM2 is critical for host defense against DNA viruses and bacteria that replicate in the cytosol, such as *Francisella tularensis* subspecies *novicida*" (PMID: 25774716), with guanylate-binding proteins (GBP2, GBP5) promoting bacteriolysis to expose ligands. The signaling axis: "AIM2, an inflammasome receptor sensing cytosolic DNA, activates caspase-1 in an ASC-dependent manner, leading to both pyroptosis and release of the proinflammatory cytokines IL-1 β and IL-18" (PMID: 23975862). Gasdermin-D executes pyroptosis and is required for host protection against *Francisella* (PMID: 30404813). An ASC-dependent, caspase-1-independent (caspase-8) pathway also generates IL-18, driving NK/T-cell IFN- γ .

F011 — Anatomical involvement centers on the reticuloendothelial system; no human causal genes

Typhoidal tularemia targets the mononuclear phagocyte/reticuloendothelial system: spleen (UBERON:0002106), liver (UBERON:0002107), lymph nodes (UBERON:0000029), bone marrow (UBERON:0002371), and lung (UBERON:0002048). The mouse data confirm "decreased bacterial burden in the spleen and liver" as the readout of dissemination (PMID: 27714591). Target cells are macrophages (CL:0000235) and dendritic cells (CL:0000451); subcellular compartments are the phagosome/late endosome (GO:0045335) and cytosol (GO:0005829). Because the disease is "caused by *Francisella tularensis*" (PMID: 40107886) — an infectious zoonosis — there are **NO human causal or susceptibility genes, no inheritance pattern, no pathogenic variants, and no genetic/carrier testing**. These genetic sections are Not Applicable; host resistance is polygenic/innate (AIM2, GBPs, IFN- γ) rather than Mendelian.

F012 — Etiology and risk factors: zoonotic/environmental exposure; immune-evasive LPS as upstream virulence trait

Risk factors are exposures, not host genotype. "The highest risk of tick-borne infection is particularly connected with people either resting or working in the forest or meadow surroundings (i.e., foresters, farmers, hunters)" (PMID: 27044720). Systemic presentation correlates with age — "systemic disease occurred more commonly in older patients" (PMID: 22911645). A key upstream virulence trait is the unusual LPS: "Modifications of the lipid A structure to less-acylated forms have been observed in some bacterial species, and those forms are poor stimulators of the TLR4/MD-2 complex" (PMID: 23745121), enabling *F. tularensis* to evade early innate recognition.

F013 — Temporal course, prognosis, and differential diagnosis

Onset is acute/subacute after a ~3–5 day incubation. Untreated systemic disease can progress to severe sepsis, respiratory failure, and death. With early appropriate antibiotics prognosis is good, but treatment failure is common when therapy is delayed: in a Turkish multicenter series of 1034 patients, "treatment failure was considered to have occurred in 495 patients (48%)" (mean 26.8 days to appropriate therapy in a predominantly oropharyngeal cohort) (PMID: 24975504). Relapse occurs with bacteriostatic agents — "one disease relapse occurred with doxycycline treatment" (PMID: 38294118). Differentials include typhoid fever, sepsis, endocarditis, brucellosis, Q fever, leptospirosis, plague, disseminated TB, EBV/CMV, and malignancy/lymphoma — the granulomas "may mimic lung cancer or lymphoma, often resulting in delayed diagnosis and unnecessary invasive investigations" (PMID: 41482246).

F014 — Synthesis

Integrating all findings, typhoidal tularemia (MONDO:0000321) is the systemic septicemic form of *F. tularensis* infection with the highest case-fatality, a purely infectious etiology (human genetic sections Not Applicable), a well-defined intracellular pathogenesis (LPS evasion → phagosomal escape → cytosolic replication → AIM2/IFN- γ defense → reticuloendothelial dissemination), serology-based diagnosis, gentamicin-first treatment, and prevention through exposure avoidance and post-exposure prophylaxis in the absence of a licensed vaccine.

Section-by-Section Characterization

1. Disease Information

- **Overview:** Severe systemic (typhoid-like) form of tularemia; febrile illness without ulcer or lymphadenopathy (F001).
- **Identifiers:** MONDO:0000321; MeSH "Tularemia" (D014406); ICD-10 A21.9 (A21 tularemia; A21.7 generalized/typhoidal); ICD-11 1B94. OMIM/Orphanet: not a genetic disease — no OMIM entry; not an Orphanet rare genetic disorder.
- **Synonyms:** Typhoidal tularemia, septicemic tularemia, systemic tularemia; ("rabbit fever," "deer-fly fever," Francis disease, Ohara disease refer to tularemia broadly).
- **Data source:** Aggregated disease-level resources plus individual case reports (EHR-derived case series).

2. Etiology

- **Causal factor:** Infectious — *Francisella tularensis* (NCBI:txid263); type A (subsp. *tularensis*, txid119856) more virulent than type B (subsp. *holarctica*, txid119857) (F002, F012).
- **Genetic risk factors:** None (no Mendelian susceptibility). Host resistance is innate/polygenic (AIM2, GBPs, IFN- γ) (F011).
- **Environmental risk factors:** Handling infected lagomorphs/rodents, hunting, farming, landscaping/mowing, lab work, arthropod bites (ticks, mosquitoes, deerflies), contaminated water/undercooked meat, aerosol inhalation; older age, male predominance (~65%) (F012).
- **Protective factors:** Prior LVS vaccination (partial); prompt antibiotics. No known protective genetic alleles.

- **Gene-environment interactions:** Not applicable in the Mendelian sense; innate-immune capacity modulates outcome.

3. Phenotypes

Symptoms/signs and lab abnormalities detailed in F009 with HPO mappings. Onset adult-predominant; severity moderate–severe; progression acute/progressive if untreated; typhoidal form ~10–14% frequency among cases (F005). Quality-of-life impact: acute severe febrile illness with prostration; full recovery expected after cure (F013).

4. Genetic/Molecular Information

Not Applicable — no human causal genes, pathogenic variants, modifier genes, epigenetic disease drivers, or chromosomal abnormalities. This is an infectious disease (F011). Relevant *bacterial* virulence loci: FPI genes *iglC/iglD*, regulators *mglA/sspA* (F003).

5. Environmental Information

Infectious agent: *F. tularensis* (F005, F012). Environmental reservoirs: terrestrial (lagomorphs/ticks) and aquatic (rodents/water). Occupational/recreational outdoor exposure is the dominant risk (F012).

6. Mechanism / Pathophysiology — Ordered Causal Chain

1. Exposure to *F. tularensis* (inhalation of ≤ 10 organisms, ingestion, bite, or contact) --leads to-->
 2. Under-acylated/hypo-phosphorylated lipid A poorly stimulates TLR4/MD-2, evading early innate recognition (<https://pubmed.ncbi.nlm.nih.gov/23745121/> rel="noopener noreferrer" title="Visit")
 3. Uptake by macrophages/dendritic cells (CL:0000235/CL:0000451) --leads to-->
 4. Francisella-containing phagosome matures to LAMP-1+ late endosome but AVOIDS lysosomal fusion; FPI IgLC/IgLD + MglA/SspA disrupt the membrane ([https://pubmed.ncbi.nlm.nih.gov/15953029/](https://pubmed.ncbi.nlm.nih.gov/15953029/ "Visit") rel="noopener noreferrer" title="Visit")
 5. Escape into the cytosol (GO:0005829) and rapid intracellular replication --leads to-->
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| <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p>BRANCH A (host defense)</p> <p>Bacteriolysis releases dsDNA;
GBP2/GBP5 expose ligands ->
AIM2 + ASC + caspase-1 ->
pyroptosis (gasdermin-D) +
IL-1b/IL-18 (https://pubmed.ncbi.nlm.nih.gov/25774716/ rel="noopener noreferrer" title="Visit")
IL-18 -> NK/T-cell IFN-g ->
macrophage activation, NO,
phagosome acidification,
bacterial restriction
(https://pubmed.ncbi.nlm.nih.gov/21700740/ rel="noopener noreferrer" title="Visit")</p> </div> <div style="width: 45%;"> <p>BRANCH B (dissemination)</p> <p>Infected phagocytes carry bacteria via
blood/lymph to reticuloendothelial organs:
spleen, liver, lymph nodes, bone marrow,
lung (https://pubmed.ncbi.nlm.nih.gov/27714591/ rel="noopener noreferrer" title="Visit")
cytopenias, hepatosplenomegaly, sepsis
(https://pubmed.ncbi.nlm.nih.gov/41458397/ rel="noopener noreferrer" title="Visit")
Systemic febrile illness = TYPHOIDAL
TULAREMIA; up to 60% fatal if untreated</p> </div> </div> |
| +-----+ |

- **Upstream steps:** LPS immune evasion, phagosomal escape (inferred from LVS/*novicida* models; *demonstrated in vitro/in vivo* in mice and arthropod cells).
- **Downstream steps:** Inflammasome activation, cytokine response, reticuloendothelial dissemination, organ injury.
- **GO terms:** phagosome maturation (GO:0090382), inflammasome complex (GO:0061702), pyroptosis (GO:0070269), positive regulation of IFN- γ production (GO:0032729).
- **CL terms:** macrophage (CL:0000235), dendritic cell (CL:0000451), NK cell (CL:0000623).

7. Anatomical Structures Affected

Primary: spleen (UBERON:0002106), liver (UBERON:0002107), lymph nodes (UBERON:0000029), bone marrow (UBERON:0002371), lung (UBERON:0002048). Secondary/complications: pericardium/heart (UBERON:0002348/UBERON:0000948), pleura (UBERON:0000977), GI tract (UBERON:0001555), kidney. Tissue: mononuclear phagocyte/reticuloendothelial system. Subcellular: phagosome/late endosome (GO:0045335), cytosol (GO:0005829). Lateralization: systemic/bilateral (F011).

8. Temporal Development

Acute/subacute onset after ~3–5 day incubation (range 1–14). Progression rapid if untreated → sepsis/respiratory failure. Self-limited to fatal depending on subspecies and treatment timing; not chronic/relapsing except with bacteriostatic therapy. Critical intervention window: early antibiotics (F013).

9. Inheritance and Population

No inheritance (infectious). Epidemiology: Northern Hemisphere zoonosis; type A (North America) vs type B (Europe/Asia); typhoidal ~10–14% of cases; male predominance ~65%; outbreaks (Sweden 2019: 979 cases). >250 animal host species (F005, F008, F012).

10. Diagnostics

Serology (microagglutination > ELISA; ICT 98.3% sens/96.5% spec), blood culture (positive more often in typhoidal disease, ~3.9% overall), PCR, cell-free DNA/metagenomic sequencing; culture requires BSL-3. Imaging (PET/CT) shows hypermetabolic granulomas mimicking malignancy. Differential diagnosis per F013 (F006, F009, F013).

11. Outcome/Prognosis

Untreated case-fatality up to 60% (typhoidal/pneumonic); with fluoroquinolone therapy in severe type B, 30-day mortality 1.5%. Poor-prognosis factors: type A (esp. type A-east), bacteremia, older age, comorbidity, delayed treatment. Recovery usually complete; no typical long-term disability (F002, F004, F013).

12. Treatment

First-line for severe/typhoidal: **gentamicin** (aminoglycoside; NCIT gentamicin C557). Alternatives: fluoroquinolones (ciprofloxacin NCIT C2669, levofloxacin) and tetracyclines (doxycycline NCIT C513; bacteriostatic, higher relapse). Streptomycin historically first-line. No pharmacogenomic guidance applicable. Experimental: novel rifampicin derivative TPR1 ± doxycycline in murine type A models ([PMID: 34223120](#)) (F004).

13. Prevention

Primary: exposure avoidance, vector/animal control, PPE, safe water/food. Immunization: no licensed vaccine (LVS partial/investigational). Secondary: prompt recognition and treatment. Tertiary: appropriate antibiotic selection to prevent relapse/complications. Post-exposure prophylaxis (doxycycline/ciprofloxacin) after recognized high-risk/aerosol exposure (F007).

14. Other Species / Natural Disease

Zoonosis affecting >250 species (NCBI:txid263). Reservoirs: cottontail rabbits (*Sylvilagus*), hares (*Lepus*), muskrats, beaver, voles; vectors: ticks, mosquitoes, deerflies. High zoonotic potential; cross-species susceptibility broad. Veterinary relevance in lagomorphs/rodents; cats can transmit to humans (F005, F008).

15. Model Organisms

Mammalian: mice (intranasal/intradermal SchuS4 [virulent] or LVS [attenuated]) recapitulate lung/spleen/liver dissemination and cytokine responses; Fischer 344 rats. Invertebrate: *Drosophila melanogaster* (arthropod-vector model of intracellular trafficking). Cellular/in vitro: macrophages, S2 cells. Bacterial mutants (*iglC*, *iglD*, *mglA*, *galU*, *clpB*) dissect virulence. Phenotype recapitulation strong for dissemination/immunity; limitation: mouse LVS is hyper-susceptible vs human, and subspecies virulence differences complicate translation (F003, F008).

Mechanistic Model / Interpretation

The central logic of typhoidal tularemia is **intracellular parasitism of the reticuloendothelial system with delayed innate recognition**. The disease's severity flows from three interlocking traits:

Trait	Molecular basis	Consequence
Immune stealth	Under-acylated, hypo-phosphorylated lipid A (PMID: 23745121)	Weak TLR4/MD-2 signaling → delayed inflammation → unchecked early replication
Phagosomal escape	FPI IgIC/IgID + MglA/SspA (PMID: 15953029 , PMID: 18474647)	Cytosolic access, avoids lysosomal killing, exponential intracellular growth
Systemic tropism	Macrophage/DC hijacking + hematogenous spread	Multi-organ seeding (spleen, liver, marrow, lung) → cytopenias, sepsis

The host's decisive countermeasure is the **AIM2 inflammasome–IFN- γ axis**. Cytosolic bacterial DNA — the very consequence of successful escape — becomes the trigger for AIM2/ASC/caspase-1 assembly, pyroptosis, and IL-1 β /IL-18 release, with IL-18 driving IFN- γ that re-activates macrophages to acidify phagosomes and restrict growth ([PMID: 25774716](#), [PMID: 23975862](#), [PMID: 21700740](#)). Clinical outcome hinges on whether this response, aided by timely bactericidal antibiotics, contains dissemination before organ failure. This explains why **early gentamicin** is decisive and why **delayed diagnosis** — driven by the non-localizing, malignancy-mimicking presentation — is the strongest modifiable determinant of mortality.

Evidence Base

PMID	Topic (abbrev.)	Supports	Evidence type
40107886	Tularemia for clinicians (review)	F001, F002, F004, F011, F014	Human clinical review
41458397	Typhoidal tularemia w/ pancytopenia & pericarditis	F001, F009	Human case report
16836829	US tularemia molecular epidemiology 1964–2004	F002	Human epidemiology
15953029	IglC/MglA phagosome biogenesis	F003	In vitro/model
18474647	MglA/Igl proteins in murine macrophages	F003	Model organism
21700740	Nitric oxide inhibits mglA, phagosomal containment	F003, F010	In vitro
38294118	Fluoroquinolones for severe type B tularemia	F004, F013	Human clinical cohort
1305858	Ecology of tularaemia	F005, F008	Review/ecology
31640844	Large Swedish outbreak 2019	F005	Human epidemiology
24874046	Norway 2011 surveillance	F005, F006	Human epidemiology
20220165	Immunochromatographic serodiagnosis	F006	Diagnostic validation
41160772	cfDNA sequencing diagnoses typhoidal case	F006	Human case report
32989563	Tularemia re-emerging (review)	F007	Review
11386933	Tularemia as biological weapon	F007	Consensus guideline
28372827	Pulmonary CD4 T cells, SchuS4 model	F008	Model organism
27714591	MAPK modulation; spleen/liver burden	F008, F011	Model organism
41482246		F009, F013	Human case series

PMID	Topic (abbrev.)	Supports	Evidence type
	Pulmonary tularemia mimicking malignancy		
25774716	GBPs promote AIM2 activation	F010	Model/in vitro
23975862	ASC/IL-18/IFN- γ in <i>F. novicida</i>	F010	Model organism
30404813	Gasdermin-D promotes AIM2, host protection	F010	Model organism
27044720	Tick-borne diseases risk (Poland)	F012	Epidemiology
22911645	Missouri 121-case review	F012	Human case series
23745121	LPS/lipid A immune evasion	F012	Review/ mechanistic
24975504	Turkish multicenter 1034 cases	F013	Human cohort
34223120	TPR1 rifampicin derivative	\$12	Model organism

Note on one citation: [PMID: 30404813](#) was flagged as a snippet "mismatch" during verification (title-level quote), while the corroborating AIM2 mechanism citations ([PMID: 25774716](#), [PMID: 23975862](#)) were verified; the gasdermin-D claim should be treated as strongly supported but with that caveat.

Limitations and Knowledge Gaps

- 1. Mechanistic studies use surrogates.** Much intracellular-trafficking and inflammasome data derive from attenuated strains (LVS) or *F. novicida* and murine/*Drosophila* models, not virulent human type A SchuS4 in humans. Direct human *in vivo* mechanistic data are scarce for biosafety reasons.
- 2. Typhoidal-specific data are limited.** The typhoidal form is uncommon (~10–14% of cases), so most epidemiologic and treatment-outcome data pool all forms; form-specific mortality figures rest largely on historical and review-level estimates.

3. **No genetic dimension.** Sections 4 and much of 9 (inheritance, variants, carrier screening) are Not Applicable; human genetic susceptibility is essentially uncharacterized beyond innate-immune pathway inference.
4. **One citation caveat.** The gasdermin-D snippet
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was a title-level mismatch during verification.
5. **Diagnostic delay is systemic.** The malignancy-mimicking, non-localizing presentation means published typhoidal cases are biased toward severe/complicated or incidentally-discovered patients.
6. **Vaccine evidence gap.** No licensed vaccine; LVS efficacy correlates remain incompletely defined.

Proposed Follow-up Experiments / Actions

1. **Form-stratified outcome meta-analysis.** Pool national surveillance datasets (Sweden, Norway, US CDC, Turkey) to derive typhoidal-specific case-fatality, time-to-treatment, and relapse rates with confidence intervals.
2. **Human AIM2/IFN- γ correlates.** Prospectively measure serum IL-18, IFN- γ , and IL-1 β in confirmed typhoidal cases to test whether inflammasome activation predicts severity/outcome.
3. **Rapid point-of-care diagnostics validation.** Evaluate cfDNA/metagenomic and multiplex-PCR assays specifically in bacteremic/typhoidal patients to shorten the ~2-week serology window.
4. **Antibiotic head-to-head.** Design a prospective registry comparing gentamicin vs fluoroquinolone vs combination therapy in severe/systemic disease, capturing relapse.
5. **Vaccine correlate studies.** Extend multifunctional T-cell/IL-17 correlate work (LVS) toward a licensable defined live-attenuated or subunit vaccine, tested against systemic challenge in the SchuS4 model.
6. **Host-directed adjuncts.** Test IFN- γ or nitric-oxide-pathway augmentation as adjuncts to antibiotics in murine systemic models, building on the mglA-inhibition finding
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Report compiled from 14 confirmed findings and 67 reviewed papers across 5 iterations. All mechanistic and clinical claims are anchored to primary literature with PMID-linked abstract quotes, except the single noted caveat.

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